

Plastics Converting Preliminary Product Review Meeting Outline

12 May 2004

Title:

Additives in Plastics Processing (Converting into Finished Products) - Generic Scenario for Estimating Occupational Exposures and Environmental Releases

This document presents a standard approach for estimating worker exposures to and environmental releases of additives used in the converting of plastic resins into finished products. It does not cover compounding plastic resins or the manufacture of polymers.

Industry Sector Description:

The plastic manufacturing industry can be divided into four sections: polymer manufacturing, compounding, converting, and “in-house” manufacturing. This generic scenario will address converting operations. Polymer manufacturing will not be included in this scenario. Compounders produce masterbatches of plastic resins with specific properties by blending the polymer (resin), additives, fillers, and reinforcements. Converters receive the masterbatch of plastic resin from compounders and form finished plastic products. Compounding and converting may take place at the same facility (“in-house” manufacturing) or at separate facilities. This scenario assumes that compounding and converting take place at separate facilities; therefore, in-house manufacturing is not covered in this scenario.

There are two types of polymers:

- *Thermoplastic polymers* are melted and become fluid when heat and pressure are applied. The molten polymers are formed into a shape via pressure. Thermoplastics solidify when cooled and the heating and cooling process can be repeated many times with little loss in properties. This generic scenario focuses on thermoplastic polymers.
- *Thermosetting polymers* (e.g., foam, epoxy) are formed into finishing products during a chemical reaction under pressure and heat. This process creates permanent cross-linking and the product retains its shape during subsequent cooling and heating. Thermosetting polymers are not within the scope of this generic scenario and are not discussed further.

NAICS code 3261, plastic product manufacturing, describes the plastic product manufacturing sector as an industry group comprised of establishments primarily engaged in processing new or spent (recycled) plastic resins into intermediate or final products using processes such as compression molding, extrusion molding, injection molding, blow molding, and casting. This industry category also includes foam production. Since foam is produced with thermosetting polymers, NAICS code 3261 sub-categories for foam are excluded when determining the size of the converting plastic

industry. Table 1 presents the 2001 County Business Patterns Data for NAICS Codes 3261.

Table 1. County Business Patterns 2001 Data for NAICS Code 3261

NAICS Code	Description	Number of Establishments	Total Number of Workers
3261	<i>Plastic Product Manufacturing</i>	13,367	803,846
32611	Unsupported plastics; film, sheet, and bag manufacturing	1,440	108,464
32612	Plastic pipe, pipe fitting, and unsupported profile shape manufacturing	1,125	49,083
32613	Laminated plastics; plate, sheet, and shape manufacturing	417	16,695
32614*	Polystyrene foam production*	566	31,503
32615*	Foam product (except polystyrene) manufacturing*	610	35,645
32616	Plastic bottle manufacturing	459	35,655
32619	Other plastic product manufacturing	8,750	526,801
Total (excluding codes 32614 and 32615)		12,191	736,698

* Foam manufacturing is not within the scope of this generic scenario.

Table 2 presents the types of thermoplastic resins, common uses, and 2003 production volume.

Table 2: Type of Thermoplastics, North American Production, And Common Uses

Resin	Production		Common Uses
	millions of pounds, dry weight basis	millions of kg, dry weight basis	
Low density polyethylene (LDPE) (a)	7,846	3,566	Packaging films, plastic bags, molded housewares, toys, containers, pipes, drums, gasoline tanks, coatings
Linear low density polyethylene (LLDPE) (a)	11,174	5,079	
High density polyethylene (HDPE) (a)	15,691	7,132	
Polypropylene (PP) (a)	17,559	7,981	Packaging, automotive, appliances, carpeting
Acrylonitrile butadiene (ABS) (b)	1,215	552	Appliances, automotive parts, pipe, business machines, telephone components
Nylon (b)	1,282	583	Automotive parts, electronics, packaging

Resin	Production		Common Uses
	millions of pounds, dry weight basis	millions of kg, dry weight basis	
Polystyrene (a)	6,442	2,928	Packaging, food service products, automotive parts, toys, housewares
Styrene Acrylonitrile (SAN) (b)	137	62	Automotive instrument panels and interior trim, housewares
Other sterics (b)	1,597	726	NA
Polyvinyl Chloride (PVC) (a)	14,696	6,680	Upholstery, flooring, wall coverings, pipe, siding, apparel
Thermoplastic polyester (b)	7,587	3,449	Plastic bottles, textiles, X-ray film, magnetic tape, packaging, metallized film, strapping, labels
Total Selected Thermoplastics	85,226	38,738	

Sources: The American Plastics Council and The Society of Plastic Engineering.

(a) Canadian production included.

(b) Canadian and Mexican production included.

NA – Not available.

Sources:

OECD 2003. Emission Scenario Document on Plastic Additives. Organisation for Economic Co-operation and Development. OECD Environmental Health and Safety Publications. July 2003.

Kirk-Othmer Encyclopedia of Chemical Technology. Plastics Processing, Plasticizers, Flame Retardants. 4th ed., 1991.

U.S. Census. County Business Patterns. U.S. Census Bureau. 2001.

<http://censtats.census.gov/cbpnaic/cbpnaic.shtml> (last confirmed December 2003)

The American Plastics Council 2003. APC Year End Statistics for 2003.

www.americanplasticscouncil.org/benefits/economics/economics.html (last confirmed March 2004)

The Society of Plastics Industry. 2003. Definitions of Plastic Resins.

www.plasticsindustry.org/industry/defs.htm (last confirmed February 2003)

Process Description:

Polymer resin is received at the compounding sites from the resin manufacturer in the form of pellets. A compounding site blends the resin and additives to produce a masterbatch. The converting site then processes the masterbatch by shaping the plastic into the desired form for the final plastic product. The blending and forming may take place at the same facility (“in house” manufacturing) or separate facilities. As a

conservative estimate, it is assumed that the compounding of the plastic resin and the converting of the resin into plastic products take place at separate facilities. Therefore, in-house manufacturing is not covered in this scenario. After shaping, finishing operations such as filing, grinding, sanding, polishing, painting, bonding, coating, engraving etc. are performed to complete the finished plastic product. This scenario covers the converting of plastic resins into finished products.

Chemicals Covered

Additives are typically added to polymer resins to adjust the properties of the plastic resin. Additive components include antioxidants, antistatics, blowing agents, colorants, coupling agents, fillers, flame retardants, heat stabilizers, impact modifiers, lubricants, plasticizers, preservatives, slip promoters, and ultraviolet stabilizers. A search of CEB's PMN database resulted in almost 70 plastic additive cases, including most of the additive types mentioned above. Table 3 presents the additive type, function, types of chemicals, physical state, and default weight fraction of additive in plastic resin. Table 4 presents typical weight fractions of additives in specific types of plastics resins.

Table 3. Types of Additives Used in Plastic Processing

Additive Type	Function	Types of Chemicals	Typical Physical State	Default Fraction In Plastic Resin
Antioxidants	Inhibit the oxidation of plastic materials that are exposed to oxygen or air at normal or high temperatures.	Alkylated phenols, amines, organic phosphites and phosphates, esters	Solid powder or pellets (a)	0.005 (a)
Antistatics	Impart a minimal to moderate degree of electrical conductivity to the plastic compound, preventing electrostatic charge accumulation on the finished product.	Quaternary ammonium compounds, anionics, amines	Solid (a)	0.01 (a)
Blowing agents	Releases gas during heating to produce a cellular form of plastic (typically used in thermosetting resins) (a)	Carbon dioxide, acetone, sodium bicarbonate, organic nitrogen compounds (a)	Solids, volatile liquids, compressed liquids, gases (a)	0.06 (a)
Colorants	Impart color to the plastic resin.	Titanium dioxides, iron oxides, anthraquinones, carbon black	Fine powder (pigments) and liquid or waxy solid (dyes) (a)	0.2 (a)
Coupling Agent	Interface between filler and plastic, bonding with both phases to improve interfacial adhesion. Typically introduced during the treatment stage of filler manufacture. (a)	Organometallic compounds, silanes (a)	Low viscosity liquid or low melting point solid (a)	0.03 (a)
Curing agent	Assist in the curing of thermosetting materials (a)	Peroxides, amines, organotin compounds (a)	Solids and liquids	0.03 (a)
Fillers	Inert materials which reduce polymer cost, improve processability, and improve mechanical properties	NA	Solid (fiber or powder)	0.55 (a)

Additive Type	Function	Types of Chemicals	Typical Physical State	Default Fraction In Plastic Resin
Flame Retardants	Reduce the tendency of the plastic product to burn.	Antimony trioxide, chlorinated paraffins, bromophenols	Solid (a)	0.4 (a)
Heat Stabilizers	Assist in maintaining the chemical and physical properties of the plastic by protecting it from the effects of heat such as color changes, undesirable surface changes, and decreases in electrical and mechanical properties.	Lead, barium-cadmium, tin, calcium zinc	Solid (sometimes liquid) (a)	0.05 (a)
Impact Modifiers	Prevent brittleness and increase the resistance of the plastic to cracking.	Natural rubber, acrylonitrile and ethylene as copolymers	Granular solid (a)	NA
Lubricants	Assist in easing the flow of the plastic in the molding and extruding processes by lubricating the metal surfaces that come into contact with the plastic.	Stearic acid, waxes, fatty acid esters, fatty acid amines	Waxy solid; soft powder (a)	0.012 (a)
Plasticizers	Increase the plastic product's flexibility and workability.	Adipates, azelates, trimellitates, DOP/DIOP/DIDP	Liquids or waxy solids (i.e., low melting point) (a)	0.5 (a)
Preservatives	Protects against fungi and bacteria	Organotin, organomercury compounds (a)	Solid (a)	NA
Slip promoters	Improve surface lubrication during processing and use (a)	Ca and Zn stearates, waxes and fatty acid amines or esters (a)	Soft powders or waxy solids (a)	0.25 (a)
Ultraviolet Stabilizers	Absorb or screen out ultra-violet radiation thereby preventing the premature degradation of the plastic product.	Benzophenones, benzotriazole, salicates	Solids (a)	0.05 (a)

Source: Kirk-Othmer

(a) Information was obtained from Emission Scenario Document on Plastic Additives. July 2003. High end of range of weight fraction is presented as default value for more specific information see Table 3.

Table 3. Fraction of Plastic Additives in Plastics Resins

Additive	Typical Fraction of Additive in Each Type of Plastic												
	LBDE	HBPE	Poly-propylene	Rigid PVC	Flexible PVC	Poly-styrene	Expanded Poly-styrene	ABS	PET	Poly-amides	Acrylics	Acetals	Poly-carbonate
Antioxidant	0.001	0.0025	0.005	0.002	-	-	-	-	-	-	-	-	-
Antistatic	0.001	0.003	-	-	-	-	-	-	-	-	-	0.01	-
Blowing agent	0.04	-	-	0.04	-	-	0.06	-	-	-	-	-	0.04
Colorants	0.03	0.03	0.03	0.005-0.03	0.02	0.01	-	0.05	0.03	0.2	0.01	0.02	0.03
Coupling agents	-	-	-	-	-	-	-	-	-	0.005	0.005	0.005	0.005
Curing agent	0.03	-	0.005	-	-	-	-	-	-	-	-	-	-
Fillers	-	-	0.4	0.1	0.1-0.5	-	-		0.55	0.4	0.35	0.3	0.4
Flame Retardant	0.2	0.2	0.04-0.4	-	0.05	0.2	0.15	0.2	0.15	0.15-0.25	-	-	-
Heat Stabilizers	-	-	-	0.01-0.05	0.02-0.05	-	-	0.0075-0.01	-	0.005	-	0.002	0.0015-0.0025
Lubricants	-	-	-	0.01	0.001-0.003	0.012	-	-	-	-	-	-	0.012
Plasticizers	-	-	-	-	0.3-0.5	-	-		0.05	0.01			
Slip promoters	0.0005	-	-	-	-	-	-	-	-	-	-	0.25	-
UV Stabilizers	-	-	0.05	0.005	-	0.005	-	0.005	0.005	0.03	0.002	0.03	0.003

Source: Emission Scenario Document on Plastic Additives. July 2003.

LBDE – Low density polyethylene

HPDE – High density polyethylene

PVC – Polyvinyl chloride

ABS – Acrylonitrile butadiene

PET – Polyethene terephthalate

Converting Operations (Forming and Molding/Shaping)

Converters and other end users receive the compounded plastic resin and shape and mold the resin into the final article. Most plastic resins are received at the converting site in solid form (e.g., pellets, sheets, films). The plastic resin is received and unloaded from the transport container (Release 1, Exposure A). Solid plastic resins must be heated and melted to shape the material into the final product. Plastic resins are heated and formed via extrusion, thermoforming, compression molding, calendering, and encapsulation. After heating and forming, the plastic may be further processed and molded into the finished product. Dust may be generated during forming processes (Release 2). Dust is expected to settle out of the air and be disposed of via solid waste or washed down with water. Molding processes include injection molding, transfer molding, compression molding, blow molding, and rotational molding. Table 5 describes the different shaping methods.

Table 5. Description of Plastic Processing

Forming Process	Description	Typical Uses
Injection molding	Plastic granules or pellets are heated and homogenized in a cylinder (usually by extrusion). By pressure the resin is injected into a cold mold where the plastic takes the shape of the mold as it solidifies.	30% of thermoplastics are processed via injection molding (a)
Extrusion	Plastic pellets or granules are heated, fluidized, homogenized, and formed continuously as the extrusion machine feeds them through a die. Immediately after the die, the material is quenched. The result is a very long plastic shape (e.g., tube, pipe, sheet, or coated wire).	Pipes, profiles, sheet, films, and wire coating (a)
Blow molding	A plastic forming process in which air is used to stretch and form plastic materials.	Packaging, bottles, containers, thin walled articles (a)
Thermoforming	Heat and pressure are applied to plastic sheets that are placed over molds and formed into various shapes.	
Rotational molding	Finely ground plastic powders are heated in a rotating mold to the point of melting and/or fusion. The melted resin evenly coats the inner surface of the rotating mold.	
Compression molding	Plastic powder or preformed plastic part is plugged into a mold cavity and compressed with pressure and heat until it takes the shape of the cavity.	Rarely used for thermoplastics (a)
Transfer Molding	The plastic is liquefied in one chamber and then injected into a closed mold cavity by means of a hydraulically operated plunger.	

Forming Process	Description	Typical Uses
Casting	Liquid plastic is poured into a mold until it hardens and takes the shape of the mold.	
Encapsulation	An object is encased in plastic and then hardened by fusion or a chemical reaction.	
Calendering	Plastic parts are squeezed between two rolls to form a thin, continuous film.	Finely gauged flexible film (a)

Source: U.S. EPA. Profile of the Rubber and Plastics Industry. U.S. EPA Office of Compliance Sector Notebook. EPA 310-R-95-016. September 1995.

(a) OECD 2003.

Although, most plastic additives are non-volatile at ambient temperature, volatile emissions are expected during process operations at high temperatures (e.g., extrusion) (Release 3, Exposure B). After shaping and molding, the plastic is cooled. Direct or indirect contact cooling water is used to cool the plastic before finishing operations and equipment cleaning for each converting process may occur with water (Release 4, Exposure C).

Trimming and Finishing Operations

Plastic finishing operations vary depending on the type of final product is produced. Most molded plastic articles require trimming of excess plastic. Trimming is performed via filing, grinding, and sanding. Solid waste is sent to landfill or incineration and dust may be washed down with water (Release 5, Exposure D). Other finishing operations include coating, polishing, bonding, and engraving. Waste from coating operations are also expected but beyond the scope of this scenario.

End of Life Disposal

Finished plastic items have a typical expected lifetime. After use, plastic items in the consumer market are released back into the environment. Used plastic articles are disposed to land or incineration or they are recycled. According to EPA's Office of Compliance Sector Notebook for the Rubber and Plastics Industry approximately 1% of plastics are recycled. A more recent report from EPA's Office of Solid Waste estimates 5.5% of plastics is recycled. The OECD Emissions Scenario Document on Plastic Additives estimates that 8% of plastics are recycled and 17% of plastics are incinerated for thermal recovery in the UK. The end of life disposal of plastic articles is outside the scope of the scenario and this release is not included.

Sources:

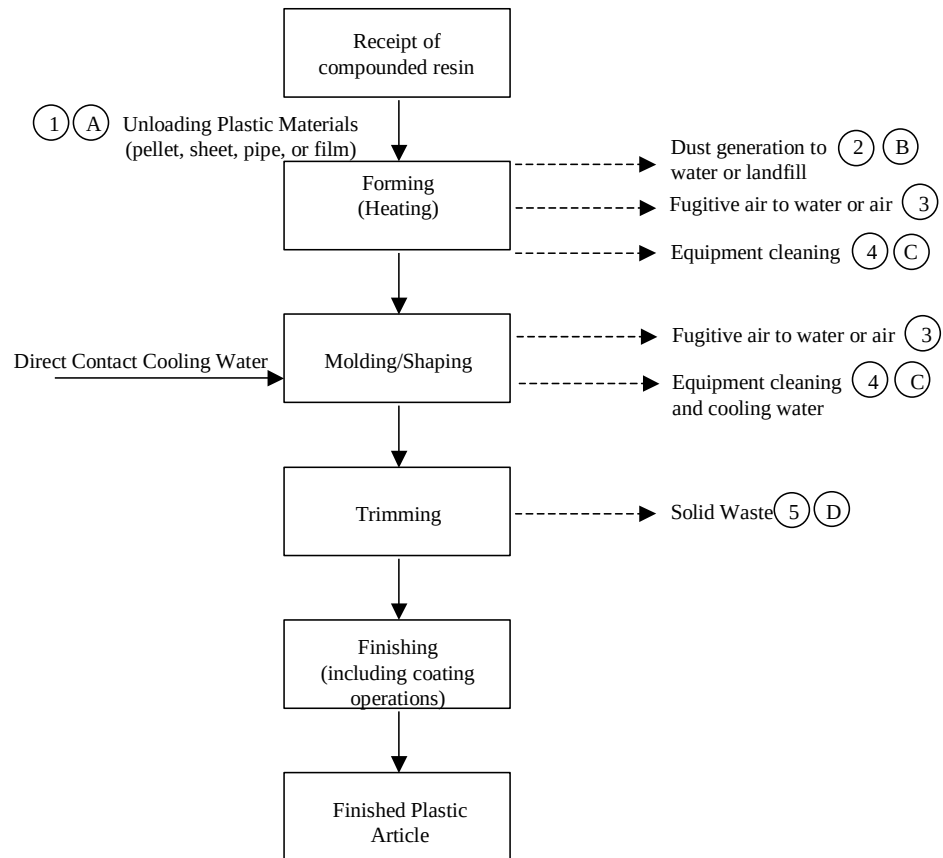
OECD 2003. Emission Scenario Document on Plastic Additives. Organisation for Economic Co-operation and Development. OECD Environmental Health and Safety Publications. July 2003.

Kirk-Othmer Encyclopedia of Chemical Technology. Plastics Processing, Plasticizers, Flame Retardants. 4th ed., 1991.

U.S. EPA. Profile of the Rubber and Plastics Industry. U.S. EPA Office of Compliance Sector Notebook. EPA 310-R-95-016. September 1995.

EPA 1984. Development Document for Effluent Limitations Guidelines and Standards for the Plastics Molding and Forming Point Source Category. Final Report. December 1984. EPA/440/1-84/069

Process Diagram:



Environmental Releases:

1. Container residue from plastic resin transport container released to water, incineration, or landfill.
2. Dust generation from forming processes released to water or landfill.
3. Fugitive air emissions from forming and molding processes released to water or air.
4. Equipment cleaning and cooling water from forming and molding processes released to water, incineration, or landfill.
5. Solid waste from trimming operations released to water or landfill.

Occupational Exposures:

- A. Exposure to solids during unloading of compounded resin from transport containers and charging to forming operation.
- B. Exposure to dusts generated from converting processes.
- C. Exposure to liquids during equipment cleaning of equipment.
- D. Exposure to solids during trimming activities.

General Facility Estimates:

I. Days of Operation

CEB standard assumption, 250 days per year based on 5 day work week and two weeks per year of operation shut down.

II. Daily Use Rate of Plastic Additive

Estimate based on the amount of thermoplastics manufactured in the U.S. (The American Plastics Council 2002, Table 2), the total number of plastic converting sites (12,191 establishment, Table 1, County Business Patterns), days of operation (CEB standard assumption), and the fraction of the additive in the thermoplastic (Table 3 or 4). This throughput will be adjusted by the percent of the chemical of interest in the additive if the additive is a mixture of chemicals. If the type of plastic is not known, the total amount of plastic resin produced in the U.S. should be used from Table 2 (38,738 million kg/yr). This estimate assumes that all converting sites (12,191 establishments) will convert all types of plastic resins.

Daily use rate = amount of resin / # converting sites / days of operation x fraction of additive x fraction of chemical in additive

Example of antioxidant in polyester resin:

5.7 kg/site-day = 3,449,000,000 kg/yr of polyester resin / 12,191 sites / 250 days/yr x 0.005

III. Number of Converting Sites

Estimate based on the total amount of plastic additive produced (PV, kg/yr); the daily use rate of plastic additive; and the days of operation. The number of sites should not exceed 12,191 converting sites.

converting sites = PV / daily use rate of additives / days of operation

Example of antioxidant in polyester resin (0.005 weight fraction, Table 3) for an assumed production volume of 100,000 kg/yr:

70 sites = 100,000 kg/yr / 5.7 kg/site-day / 250 days/yr

The values for days of operation, daily use rate of plastic additive, and number of converting sites are related. This scenario presents an equation to calculate the daily use rate of the plastic additive from the amount of resin produced, total number of sites, and days of operation with default values for each. The daily use rate and the supplied production volume are then used to calculate the number of converting sites.

If the number of converting sites is known, the daily use rate can be calculated directly without the use of the equation in section II. This alternative calculation is: daily use rate

= production volume / (number of sites x days of operation). It is recommended to calculate the daily use rate based on information on section II and compare it to the daily use rate based on number of sites.

Environmental Releases

1. Container Residue from Compounding Transport Container (Release 1)

Compounded plastic resins are shipped in bulk containers in pellet, sheet, pipe, or film form. During transportation wearing of pellets may create dust residual. Standard CEB estimate for solid container residue is 1% loss. The media of release is uncertain; therefore, container residue is assumed to be released to water, landfill, or incineration (CEB standard estimate). Specific information on size of containers is not available; therefore, 55 gal drum (208 L) is assumed as the default. 1 kg/L is assumed as the default density of the compounded resin if specific information is not available. The information presented for this standard CEB model/estimate is based on the current version of the model (as of the date of this generic scenario). Standard CEB models are subject to change; therefore, the current version of the standard CEB model should be used.

Number of Containers (containers/site-year) = Daily Use Rate (kg/site-day) x
Days of Operation (days) / Fraction of additive in resin / Container Volume
(L/container) / Density (kg/L)

Example for antioxidant transported in drums:

1,370 containers/site-yr = 5.7 kg/site-day x 250 days / (0.005 x 208 L/container x
1 kg/L)

If the number of containers is less than the days of operation, the days of release is equal to the number of containers, and the daily release is calculated based on the following formula:

Daily Release from Container Residue (kg/site-day) = Container Volume
(L/container) x Density (kg/L) x Fraction of additive in resin x Loss Fraction

Example for solid antioxidant transported in drums:

0.01 kg/site-container = 208 L/container x 1 kg/L x 0.005 x 0.01

If the number of containers is greater than the days of operation, the days of release is equal to the days of operation, and the daily release is calculated based on the following formula:

Daily Release from Container Residue (kg/site-day) = Daily Use Rate (kg/site-day) x Loss Fraction

Example of daily release of antioxidant in drums:

$$0.057 \text{ kg/site-day} = 5.7 \text{ kg/site-day} \times 0.01$$

2. Dust Generation from Converting Activities Released to Water or Landfill (Release 2)

Dust generation is expected from converting activities for solid compounded resins (0.01% loss of daily use rate, OECD 2003, based on estimates for filler additives). Particles are originally released to air but are expected to eventually settle to the ground. Dust particles are expected to be cleaned from the equipment and floor with water or disposed of directly to landfill (OECD 2003).

$$\text{Daily release of dust} = \text{daily use rate} \times \text{loss fraction}$$

Example of antioxidant in polyester resin:

$$0.00057 \text{ kg/site-day released} = 5.7 \text{ kg/site-day} \times 0.0001$$

3. Fugitive Air from Converting Activities Released to Water or Air (Release 3)

Most plastic additives are non-volatile; however, converting operations may be performed at elevated temperatures; thus slight volatilization may occur. A release of 0.25% of the daily use rate is estimated (OECD 2003, loss rates are based on high (0.25%), medium (0.05%), and low (0.01%) volatility at 200°C for typical plasticizers in open processes such as calendaring, blown film, and spread coating). Particles are originally released to air but subsequent condensation may result in losses to water. As a conservative estimate, 50% is assumed to be released to water and 50% is assumed to be released to the atmosphere (OECD 2003).

$$\text{Daily release to water from volatilization} = \text{daily use rate} \times \text{loss fraction}$$

Example of antioxidant additive:

$$0.0071 \text{ kg/site-day released to water} = 5.7 \text{ kg/site-day} \times 0.00125$$

$$\text{Daily release to air from volatilization} = \text{daily use rate} \times \text{loss fraction}$$

Example of antioxidant additive:

$$0.0071 \text{ kg/site-day released to air} = 5.7 \text{ kg/site-day} \times 0.00125$$

The table below shows the vapor pressures of common plasticizers and volatility groupings used in the OECD document.

Temperature (°C)	Vapor Pressure (hPa; 1 hPa = 0.75 mm Hg)			
	DEHA	BBP	DEHP	DIDP
20	6.3×10^{-7}	6.5×10^{-6}	2.2×10^{-7}	3.3×10^{-8}
50	2.7×10^{-5}	1.5×10^{-4}	1.0×10^{-5}	1.8×10^{-6}
100	3.7×10^{-3}	9.1×10^{-3}	1.6×10^{-3}	3.4×10^{-4}
150	0.16	0.21	0.076	0.019
200	3.1	2.5	1.6	0.46
Volatility grouping	high	high	medium	low

The table below shows the loss estimates from the OECD document for low to high volatility compounds for open and partially open/closed processes.

Process	Fugitive Air Loss	Loss to Water	Loss to Air
Open Process (calendering, blown film, spread coating)			
High volatility	0.25%	0.125%	0.125%
Medium volatility	0.05%	0.025%	0.025%
Low volatility	0.01%	0.005%	0.005%
Partially Open/Closed Process (extrusion, injection molding)			
High volatility	0.05%	0.025%	0.025%
Medium volatility	0.01%	0.005%	0.005%
Low volatility	0.002%	0.001%	0.001%

4. Residual from Converting Equipment Cleaning (Release 4)

This release includes cooling water used during converting and equipment cleaning water. The release is based on CEB standard estimate for equipment cleaning of entire process (2% loss of daily use rate) released to water. (Sector Notebook and EPA 1984 state that water is used as a direct coolant routinely during extrusion and compounding). The use of cooling water and water to clean equipment is a conservative estimate. The Development Document for Effluent Limitation Guidelines for the Plastics Molding and Forming Point Source Category (1984) estimates that 1,898 sites use process water. Process water includes contact cooling and heating water, cleaning water, and finishing water. According to past PMN cases, cleaning may also be performed by processing a “cleaning” resin through the equipment and then the “cleaning” resin is disposed. As a conservative estimate it is reasonable to assume that equipment cleaning residual is disposed of to water, incineration, or landfill.

Daily release from equipment cleaning = daily use rate x loss fraction

Example of antioxidant additive:

0.11 kg/site-day released = 5.7 kg/site-day x 0.02

5. Trimming Waste (Release 5)

Waste generated from trimming, filing, and grinding (off-spec products are reground and reintroduced into the converting process) are estimated to be 2.5-10% of production. The loss of 10% is typical of waste generated during the trimming of back-coated fabric. Back coating operations are not included in the scope of this scenario; therefore, 2.5% waste from grinding and trimming of plastic parts is representative of the industry covered in this scenario. Particles are originally released to air but are expected to eventually settle to the ground. Dust particles are expected to be cleaned from the equipment and floor with water or disposed of directly to landfill. (OECD 2003). The Development Document for Effluent Limitation Guidelines for the Plastics Molding and Forming Point Source Category (1984) indicates that finished articles are sometimes rinsed with water to remove debris and clean the finished article.

Daily release from trimmings = daily use rate x loss fraction

Example of antioxidant additive:

0.14 kg/site-day released = 5.7 kg/site-day x 0.025

Control Technologies:

Water: According to the Development Document for Effluent Limitation Guidelines for the Plastics Molding and Forming Point Source Category (1984), approximately 31% of surveyed sites that use process water directly discharged their process water; 44% indirectly discharged (POTW); and 25% had a zero discharge. Zero discharge methods include recycling, evaporation pond, septic tank with leach field, evaporation from equipment, land application, and contract haul. Types of on-site treatment include settling, pH adjustment, activated sludge, activated carbon adsorption, filtration, and vacuum filtration.

Air: The Emissions Scenario Document on Plastic Additives suggests that bag filters used to collect particulate emissions are 99% efficient. However, the prevalence of bag filter use was not available.

Occupational Exposures:

Number of Workers Exposed per Site

Estimate based on total number of workers in NAICS code 3261 excluding foam (736,698 workers, Table 1, County Business Patterns), total number of sites (12,191 establishments, Table 1, County Business Patterns), and percent of production workers in industry (79%, Annual Survey of Manufacturers).

Note, production workers includes workers (up through the linesupervisor level) engaged in fabricating, processing, assembling, inspecting, receiving, storing, handling,

packing, warehousing, shipping (but not delivering), maintenance, repair, janitorial and guard services, product development, auxiliary production for plant's own use (e.g., power plant), recordkeeping, and other services closely associated with these production operations at the establishment. All other employees covers nonproduction employees of the manufacturing establishment, including those engaged in factory supervision above the line-supervisor level, sales (including driver-salespersons), sales delivery (highway truck drivers and their helpers), advertising, credit, collection, installation and servicing of own products, clerical and routine office functions, executive, purchasing, financing, legal, personnel (including cafeteria, medical, etc.), professional, technical employees, and employees on the payroll of the manufacturing establishment engaged in the construction of major additions or alterations utilized as a separate work force.

workers/site exposed = total workers in industry / number of sites in industry x % production workers

Calculation of number of workers exposed per site:

48 workers/site exposed = 736,698 total workers / 12,191 total sites x 0.79

A. Exposure from Unloading and Charging Compounded Resin (Exposure A)

Unloading containers of compounded resin may generate dust. Compounded resin is received in pellet, sheet, pipe, or film form. Workers are involved in unloading the transport containers to the converting equipment. During this operation, workers are expected to have dermal and inhalation exposures to solids. Specific information on the number of workers involved in this activity is not available; therefore, it is assumed that at least 12 workers will be involved in the unloading and charging operations. Information regarding personal protective equipment (PPE) and engineering controls for this activity was not available.

Inhalation: CEB standard estimated based on OSHA PEL for particulates not otherwise regulated and fraction of chemical in resin

Inhalation exposure = OSHA PEL x breathing rate x hours x fraction of additive in resin x fraction of chemical in additive (if applicable)

Example of antioxidant additive:

0.75 mg/day exposure = 15 mg/m³ x 1.25 m³/hr x 8 hrs x 0.005 x 1

Dermal: Plastics are transported as pellets, sheets, films or pipes (Kirk-Othmer), which consist of large particles sizes (although some dust may be generated during transportation). Additives are incorporated into the plastic material. Some surface contact may occur; however, dermal exposure is non-quantifiable (Cast solids, CEB Method for Screening-Level Assessments of Dermal Exposure).

B. Exposure from Converting Processes (Exposure B)

Workers in the plastic processing area may be exposed to dust and vapors generated during the converting process. A NIOSH Health Hazard Evaluation Report performed air sampling of a compression molding process 1997. Inhalable dust samples were taken and results ranged from 0.29 to 1.07 mg/m³. This study indicates that dust may be present in the production area of plastic processing operations. Therefore, as a conservative estimate it is assume that workers will be exposed to dust at a maximum of the OSHA PEL for particulates not otherwise regulated (15 mg/m³). Specific information on the number of workers exposed is not available; therefore, it is assumed that at least 12 workers will be exposed to dust and vapors generated during converting processes. Information regarding PPE and engineering controls for this activity was not available.

Inhalation: CEB standard estimated based on OSHA PEL for particulates not otherwise regulated and fraction of chemical in resin

Inhalation exposure = OSHA PEL x breathing rate x hours x
fraction of additive in resin x fraction of chemical in additive (if
applicable)

Example of antioxidant additive:

0.75 mg/day exposure = 15 mg/m³ x 1.25 m³/hr x 8 hrs x 0.005 x 1

Dermal: Not expected, converting processes typically take place at high temperatures.

C. Exposure from Converting Equipment Cleaning (Exposure C)

Workers may be exposed to plastic additives during manual equipment cleaning operations. Specific information on equipment cleaning was not available; therefore, it is assumed that equipment cleaning may take place manually. Dermal exposure is expected from manual equipment cleaning. Specific information on the number workers involved in this activity is not available; therefore, it is assumed that at least 12 workers will be involved in equipment cleaning activities. Information regarding PPE and engineering controls for this activity was not available.

Inhalation: Not expected, particles are expected to be contained in water.

Dermal: CEB standard estimate for routine contact of liquids, 2 hands from equipment cleaning wastewater. Assume concentration of chemical in water is 50% the concentration of chemical in resin (engineering judgment). Note, dermal contact with cooling water is not expected because of elevated temperatures.

Dermal Exposure = liquid amount on skin x area of skin x # of incidents x fraction of chemical in additive x fraction of additive in resin x fraction of resin in water

Example for antioxidant additive:

4.4 mg/day exposure = 2.1 mg/cm²-incident x 840 cm² x 1 incident/day x 1 x 0.005 x 0.5

D. Exposure from Trimming Processes (Exposure D)

Trimming operations include sanding and grinding excess plastic material from a finished article or regrounding an off-spec product to reintroduce in the process. Workers may be exposed to dust while performing trimming operations manually. Specific information on the number workers involved in this activity is not available; therefore, it is assumed that at least 12 workers will be involved in trimming operations. Information regarding PPE and engineering controls for this activity was not available.

Inhalation: CEB standard estimated based on OSHA PEL for particulates not otherwise regulated and fraction of chemical in resin

Inhalation exposure = OSHA PEL x breathing rate x hours x fraction of additive in resin x fraction of chemical in additive (if applicable)

Example of antioxidant additive:

0.75 mg/day exposure = 15 mg/m³ x 1.25 m³/hr x 8 hrs x 0.005 x 1

Dermal: Additives are incorporated into the plastic material. Some surface contact may occur; however, dermal exposure is non-quantifiable (Cast solids, CEB Method for Screening-Level Assessments of Dermal Exposure).

Other Sources:

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AP-42 Analogous Industries:

ERG reviewed the AP-42 information for data from analogous industries. Related industry data identified includes: Plastic resin manufacturing for 1) PVC, 2) poly(ethylene Terephthalate), 3) polystyrene, and 4) polypropylene. PVC, Poly(ethylene terephthalate), and polypropylene AP-42 sections both present particulate emission factors. Although, this generic scenario does not include polymer manufacturing in the scope, the process description for the manufacture of poly(ethylene terephthalate) and polypropylene polymers include a chopper/crushing operation. This process may be similar to mixing operations for compounding. Polystyrene AP-42 section does not present particulate emission factors but does indicate in the process description an extrusion and pelletizer operation. AP-42 section on polyester resin plastic products fabrication has been removed from the EPA website stating that emission factors appear to under predict styrene emissions. The table below summarizes the available emission factors for related industries.

Emission Factors for Plastic Manufacturing Operations from AP-42

Type of plastic/operation	Particulate		Gases	
	Emission factor	Notes	Emission factor	Notes
PVC	17.5 kg/Mg	Usually controlled with fabric filter, efficiency 98-99%	8.5 kg/Mg	as vinyl chloride

Type of plastic/operation	Particulate		Gases	
	Emission factor	Notes	Emission factor	Notes
Poly(ethylene terephthalate)				
Product storage	0.0003 g/kg	0.4 g/kg if no controls	NA	
Total plant	0.17 g/kg		NA	
Polystyrene (extrusion quench vent)	NA		0.01-0.3 VOC/kg	non-methane VOC
Polypropylene	1.5 kg/Mg		0.35 kg/Mg	as propylene

Below is a comparison of PVC particulate emission factor (highest factor) with releases calculated for Releases 2 and 3.

AP-42 Emission factor for colorant additive in PVC manufacturing:
 $6,680,000,000 \text{ kg/yr of PVC} / 12,191 \text{ sites} / 250 \text{ days/yr} \times 1 \text{ Mg/1000 kg} \times 17.5 \text{ kg/Mg} \times 0.02 \text{ (fraction of additive)} = 0.77 \text{ kg/site-day released to air}$

OECD release factor of 0.0026 (combination of Release 2 and 3):
 $6,680,000,000 \text{ kg/yr of PVC} / 12,191 \text{ sites} / 250 \text{ days/yr} \times 0.02 \text{ (fraction of additive)} \times 0.0026 = 0.11 \text{ kg/site-day released to air}$

The highest emission factor from an analogous industry (manufacture of PVC) is within an order of magnitude of the OECD based estimates. This shows that estimates based on OECD data is reasonable.